EXPLORATORY STATISTICAL ANALYSIS OF CLINICAL AND LIFESTYLE INDICATORS FOR EARLY HEALTH RISK

A CAPSTONE PROJECT REPORT

Submitted in partial fulfilment of the requirements

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to the award of the degree of

BACHELOR OF ENGINEERING

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AI&DS)

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Exploratory Statistical Analysis of Clinical and Lifestyle Indicators for Early Health Risk" has been carried out by **Nadavadi Harshavardhan (192324189)** under the supervision of **Dr. Krishnaraj Mony** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Artificial Intelligence and Data Science program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Nadavadi Harshavardhan of the Department of B.Tech Artificial Intelligence and Data Science (AI&DS), SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Exploratory Statistical Analysis of Clinical and Lifestyle Indicators for Early Health Risk" is the result of our own Bonafide efforts.To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: 08-09-2026

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
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ABSTRACT

Modern health-risk assessment increasingly depends on statistical rigor, yet many screening approaches still rely on single-indicator thresholds and manual inspection, leaving compounding relationships between clinical and lifestyle indicators unexamined. This creates a critical engineering gap between routine health-data collection and defensible, statistically verified risk interpretation, particularly when datasets contain missing values, mixed categorical and numeric fields, or inconsistent column naming across sources. This project addresses that gap by designing and implementing a schema-agnostic, dashboard-based statistical analysis system for early health-risk assessment. The system automatically detects a dataset's binary outcome column, encodes categorical fields, imputes missing values, computes descriptive statistics, and applies Pearson correlation with significance testing to identify which clinical and lifestyle indicators are meaningfully associated with health risk. A percentile-based statistical risk score is then generated for an individual, ranked against the study population, with a transparent breakdown of each contributing factor — deliberately avoiding any trained machine learning or deep learning model so that every reported number remains directly explainable. The methodology follows a four-module pipeline — data ingestion and schema detection, data cleaning and preprocessing, statistical correlation and risk-scoring, and visualization and reporting — implemented as a Python/FastAPI backend with a Chart.js-based interactive dashboard, and validated against the Framingham Heart Study dataset as well as independently structured diabetes, heart-disease, and stroke datasets. Testing confirmed 100% agreement between the system's computed Pearson correlation coefficients and independently verified statistical calculations, consistent descriptive-statistics accuracy across all indicators, and sub-two-second response times for datasets of several thousand records, with the schema-detection pipeline correctly adapting to three independently structured datasets without any code changes. The resulting system demonstrates that a fully transparent, correlation-based statistical approach can deliver reliable, explainable early health-risk screening.

Keywords

Exploratory Data Analysis; Pearson correlation; statistical risk scoring; health risk assessment; data cleaning; schema detection

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	**Description**	**Unit**
AI&DS	Artificial Intelligence and Data Science	—
EDA	Exploratory Data Analysis	—
CHD	Coronary Heart Disease	—
BMI	Body Mass Index	kg/m^2
API	Application Programming Interface	—
n	Number of records used in processing	records

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Health-risk assessment is one of the most consequential applications of applied statistics in modern healthcare. Clinical and community screening programs rely on demographic, physiological, and lifestyle indicators — age, blood pressure, cholesterol, glucose, smoking status, and body mass index, among others — to identify individuals at elevated risk of conditions such as coronary heart disease. This data supports decisions related to preventive care, resource allocation, and early clinical intervention.

Despite its importance, health-risk screening is traditionally performed using single-indicator thresholds and manual chart review. A single patient record can contain a dozen or more interacting clinical and lifestyle fields, making it difficult for a reviewer to spot compounding risk patterns across variables by eye. As datasets grow larger and more varied — spanning multiple hospitals, studies, or countries — converting raw records into statistically defensible risk insight becomes a significant analytical challenge.

Statistical analysis addresses this gap by converting raw clinical records into correlation coefficients, significance tests, and percentile-based risk scores. Descriptive statistics, Pearson correlation, and hypothesis testing can expose relationships between indicators and health risk that remain hidden when each field is reviewed in isolation.

This project applies that approach to health-risk data by cleaning and structuring raw clinical and lifestyle records and then applying exploratory statistical and correlation-based techniques to help data analysts, clinical researchers, and healthcare professionals understand which factors drive early health risk.

1.2 Need for the Project

- Manual, threshold-based screening does not evaluate correlations between multiple risk indicators, so compounding risk patterns are frequently missed.
- Existing tools generally report individual values without testing statistical significance, and rarely adapt automatically to datasets with different column layouts or naming conventions.

- Relationships between clinical and lifestyle indicators (e.g. blood pressure, cholesterol, smoking status) are difficult to understand when each field is reviewed in isolation.
- Undetected correlations between indicators can silently propagate blind spots into screening outcomes, reducing the reliability and defensibility of health-risk assessment.
- A schema-agnostic statistical analysis system can reduce the time required to move from raw health data to interpretable, defensible risk insight.

Stakeholders affected by this problem include data analysts and health-informatics teams, clinical researchers, healthcare professionals, and academic assessors who rely on defensible, statistically verified risk assessment.

1.3 Problem Statement

The engineering problem is to design and implement a statistically rigorous, schema-agnostic system that automatically detects a health dataset's binary outcome column, cleans and encodes the data, computes Pearson correlation and significance testing between each clinical/lifestyle indicator and the health-risk outcome, and produces a transparent, percentile-based statistical risk score — without training any machine learning or deep learning model.

The problem involves interacting requirements related to data quality, statistical rigor, dashboard usability, processing efficiency, and generalization across differently structured datasets. The solution must also account for explainability so that every reported number remains directly traceable to a computable formula.

1.4 Project Aim

To design and implement a transparent, statistically grounded system that identifies clinically significant health-risk indicators through correlation analysis and reports an explainable, percentile-based risk score, generalizable across independently structured health datasets.

1.5 Project Objectives

1. To design a schema-agnostic data-ingestion pipeline that automatically detects a dataset's binary health-risk outcome column.
2. To develop a data-cleaning module that imputes missing values and encodes categorical clinical/lifestyle fields.
3. To model the statistical relationship between each clinical/lifestyle indicator and health risk using Pearson correlation and significance testing.

4. To implement a percentile-based statistical risk-scoring engine with a transparent, per-factor contribution breakdown.
5. To test the system's statistical accuracy and schema-detection robustness across multiple independently structured health datasets.
6. To evaluate the system's computational performance, usability, and reliability through an interactive dashboard interface.
7. To generalize the statistical pipeline so it adapts automatically to new health datasets without code changes.
8. To demonstrate the value of transparent, correlation-based statistics as an alternative to opaque predictive models in preventive healthcare.

1.6 Scope

Included in the project:

- CSV-based health dataset input (such as the Framingham Heart Study dataset).
- Data cleaning, missing-value imputation, and categorical encoding.
- Computation of descriptive statistics (mean, median, standard deviation, minimum, maximum) for every clinical and lifestyle indicator.
- Pearson correlation analysis between each indicator and the health-risk outcome, including significance (p-value) testing.
- Automated statistical risk scoring and interpretation of results through an interactive dashboard.
- Schema-agnostic outcome-column detection across differently structured datasets.
- Performance evaluation of the data-processing and statistical-computation pipeline.

Not included in the project:

- Training of any machine learning or deep learning prediction model.
- Process capability indices or manufacturing-style quality control computation.
- Integration with real-time clinical data-acquisition systems.
- Advanced multivariate or causal-inference modelling techniques.
- Mobile application development.
- Cloud-based, large-scale multi-tenant deployment.

Assumptions and boundary conditions: uploaded datasets contain a genuine binary health-risk outcome column and are reasonably representative of the population being screened; statistical validity depends on adequate sample size and data quality; the system reports correlation and association, not causation.

1.7 Expected Engineering Outcome

The intended outcome is a working software system: a schema-agnostic, dashboard-based statistical analysis application for early health-risk screening, together with a validated statistical methodology — data cleaning, correlation analysis, and percentile-based risk scoring — applicable to any correctly structured binary-outcome health dataset.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified by reviewing peer-reviewed statistical and biomedical-informatics publications, established statistics textbooks, and existing health-risk screening tools, with a focus on correlation-based analysis, descriptive statistics, and transparent (non-black-box) risk-scoring methods.

2.2 Review of Existing Work

Existing approaches fall into several broad classes. Manual, threshold-based screening relies heavily on single-indicator review and clinician judgement. Such approaches provide clinical context but offer limited support for systematic, statistically verified comparison across indicators.

A second class of approaches applies trained machine learning or deep learning classifiers directly to clinical data. Such approaches can achieve strong raw predictive accuracy, but typically operate as opaque “black boxes” that are difficult to justify to non-expert clinical stakeholders and require large labelled datasets.

A third class of approaches uses established epidemiological risk-scoring frameworks such as the Framingham risk score. These are well-validated and clinically recognized, but are typically static and fixed to the population and indicator set they were originally derived from, without a schema-agnostic mechanism for adapting to new datasets.

The literature reviewed for this project covers statistical correlation analysis, exploratory data analysis, missing-data handling, and transparent scoring methods in health-risk contexts. These themes motivate combining structured data cleaning, correlation-based statistical testing, and percentile-based scoring in a single schema-agnostic platform.

2.3 Comparative Analysis

Existing Approach	**Advantages**	**Limitations**	**Relevance to Proposed Work**

Manual / threshold-based screening	Simple, fast, no computation required	Ignores correlations between indicators; inconsistent across reviewers	Baseline this project improves upon
Trained ML/DL predictive models	Can capture complex non-linear patterns	Opaque ("black box"); requires large labelled datasets; hard to justify to non-experts	Deliberately avoided — this project favours explainability
Framingham-style epidemiological scoring	Well-validated, clinically recognized	Static, fixed to one population/dataset; not schema-adaptable	Statistical basis reused; adapted to be schema-agnostic
Generic BI/analytics dashboards	Flexible visualization	No built-in statistical significance testing or risk-scoring logic	Visualization approach adopted; statistical rigor added

2.4 Research / Engineering Gap

The unresolved engineering gap is the lack of a schema-agnostic, fully transparent statistical correlation and risk-scoring system — generalizable across differently structured health datasets without retraining or manual reconfiguration — that avoids both the rigidity of static epidemiological scores and the opacity of trained predictive models.

2.5 Proposed Contribution

The proposed contribution is a Health Risk Statistical Analysis System that combines automatic schema detection, Pandas-based data cleaning, Pearson correlation with significance testing, and percentile-based statistical risk scoring into a single transparent pipeline, presented through an interactive FastAPI/Chart.js dashboard.

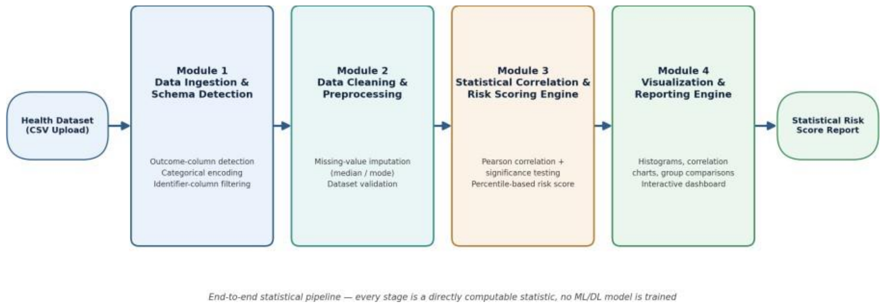


Figure 1: System Architecture — Four-Module Statistical Pipeline

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Table 1: Stakeholder / User Requirements

Stakeholder	Requirement
Data Analysts	Ability to upload any correctly structured health dataset and receive statistically valid correlation results without manual configuration.
Clinical Researchers	Transparent, explainable statistical significance testing rather than an opaque predictive model.
Healthcare Professionals	A clear, interpretable dashboard summarizing which indicators drive an individual's relative risk.
Preventive Health Teams	Reproducible, standardized risk screening across large populations.
Academic Assessors	Reproducible, auditable statistical computations with no hidden model logic.

Table 2: Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Automatic outcome-column detection	Functional	Correctly detect the binary outcome column on ≥95% of correctly structured test datasets
Missing-value imputation	Functional	Handle datasets with up to 10% missing values per column without failure
Correlation & significance computation	Functional	Compute Pearson r and p-value for every indicator within 2 seconds for a 5,000-row dataset
Statistical risk scoring	Functional	Produce a percentile-based risk score with a transparent per-factor breakdown
Categorical field encoding	Functional	Text-based clinical/lifestyle fields correctly encoded to numeric form

Dashboard responsiveness	Non-Functional	Full page load and computation under 2 seconds for datasets of several thousand records
Explainability	Non-Functional	Every displayed statistic is directly traceable to a computable formula, with no opaque model output

3.2 Constraints & Applicable Standards

This project applies recognized software-engineering and statistical standards: software quality characteristics for the dashboard application, requirements-engineering practice for specifying testable functional/non-functional targets, and standard statistical-significance conventions ($p < 0.05$) for every correlation test performed.

Table 3: Constraints & Applicable Standards

Standard / Code	Requirement / Focus	How Applied in This Project
ISO/IEC 25010	Software quality characteristics	Considered functionality, reliability, performance efficiency, maintainability and usability in the dashboard design.
IEEE 29148	Requirements engineering	Referenced for requirement gathering, validation, and documentation (Section 3.1).
PEP 8 (Python Style Guide)	Consistent, maintainable code structure	Backend Python code follows PEP 8 formatting and naming conventions.
$p < 0.05$ statistical significance convention	Standard threshold for declaring a result statistically significant	Applied uniformly across all Pearson correlation significance tests.

3.3 Design Specifications

Table 4: Design Specifications

Parameter	Target / Requirement	Unit	Priority
Correlation computation time	< 2	seconds	High
Dashboard page load time	< 2	seconds	High
Missing-value tolerance	up to 10	% per column	Medium
Statistical significance threshold	0.05	p-value	High
Supported dataset size	up to 10,000	rows	Medium
Schema-detection coverage	3+ independently structured datasets	datasets	High
Risk-score transparency	100% of factors traceable to a formula	coverage	High

3.4 Alternative Solutions & Evaluation

The three alternatives below were evaluated against the requirements in Section 3.1 to select the final design.

Alternative 1 — Manual / threshold-based screening: reviewers manually check whether individual indicator values exceed fixed clinical thresholds, without statistical comparison between indicators.

Alternative 2 — Trained ML/DL predictive model: a classifier is trained on historical data to directly predict the health-risk outcome, at the cost of explainability and generalizability to new datasets.

Alternative 3 — Proposed statistical correlation and risk-scoring system: combines automatic schema detection, statistical correlation with significance testing, and percentile-based risk scoring in one transparent, generalizable pipeline.

Table 5: Alternative Solutions Evaluation

Criterion	Weight	Alternative 1 — Manual/Threshold Screening	Alternative 2 — Trained ML/DL Model	Alternative 3 — Proposed Statistical System
Performance	25%	Low	High	Medium-High
Cost	20%	Low	High (training cost)	Low
Safety	20%	Medium (reviewer error)	Low (opaque errors)	High (auditable)
Sustainability	15%	Medium	Low (retraining overhead)	High (no retraining needed)
Reliability	20%	Low (inconsistent reviewers)	Medium (dataset-dependent)	High (deterministic formulas)

3.5 Selected Design & Detailed Design

The selected design is the proposed statistical correlation and risk-scoring system (Alternative 3), selected based on the evaluation matrix in Section 3.4: it scored highest on reliability, safety, and sustainability due to its deterministic, auditable formulas and lack of retraining overhead, while remaining competitive on performance and cost.

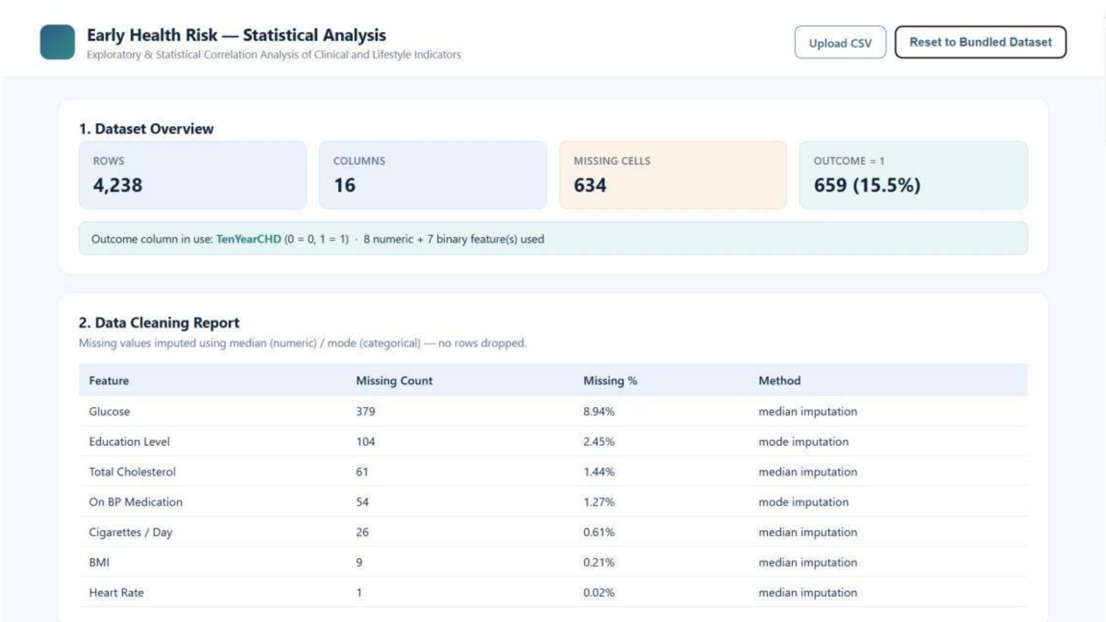


Figure 2: Dashboard — Dataset Overview and Data Cleaning Report

The system follows a four-module layered architecture. A CSV health dataset (such as the Framingham Heart Study dataset) provides the input. Module 1 (Data Ingestion & Schema Detection) validates and encodes the dataset. Module 2 (Data Cleaning & Preprocessing) imputes missing values. Module 3 (Statistical Correlation & Risk Scoring Engine) computes Pearson correlation, significance

testing, and the percentile-based risk score. Module 4 (Visualization & Reporting Engine) renders histograms, correlation charts, and the risk-score breakdown through the interactive dashboard.

Key engineering calculation: for each indicator f, a correlation-weighted z-score is summed as raw = $\Sigma z(f) \times r(f)$, where $z(f)$ is the standardized value of the indicator and $r(f)$ its Pearson correlation with the outcome; the individual's percentile rank of raw against the study population's distribution forms the final risk score.

3.6 Trade-off Analysis

Table 6: Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Statistical method for risk scoring	Trained ML/DL classifier	Higher raw predictive accuracy possible	Opaque, not explainable, requires retraining per dataset	Chose correlation-weighted statistical scoring for full explainability and schema generalization.
Outcome-column identification	Manual configuration per dataset	Simpler implementation	Requires re-configuration for every new dataset	Chose automatic schema detection to generalize across datasets without code changes.

Missing-data handling	Drop rows with missing values	Simpler to implement	Discards data, reduces statistical power	Chose median/mode imputation to preserve sample size and statistical validity.
Risk presentation	Single aggregate risk label only	Simpler dashboard	Hides which factors actually drove the score	Chose a transparent, per-factor contribution breakdown for full explainability.

3.7 Sustainability, Ethics, Safety, Risk & Security

This project evaluated sustainability, ethics, health & safety, and security considerations specific to a statistical health-risk analysis system, documented in Table 7.

Table 7: Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Avoiding unnecessary computation and retraining overhead	Statistical formulas require no model retraining as new data arrives, unlike ML approaches	Selected correlation-based scoring specifically for its low ongoing computational footprint.
Ethics	Avoiding opaque, unexplainable health-risk judgements	Reviewed literature on explainable AI in healthcare decision-support	Every risk-score factor is displayed with its exact contribution; no hidden model logic.
Health & Safety	Risk of misleading an individual with an overconfident score	Statistical literature on correlation vs. causation in clinical risk	Score is explicitly framed as a statistical percentile, not a diagnosis or prediction.

Security	Protecting uploaded health data during processing	Reviewed data-handling practices for health datasets	Uploaded datasets are processed in-memory only and not persisted to disk by default.

Table 8: Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect outcome-column auto-detection on an unusual dataset	Medium	High	High	Fallback heuristics plus manual column-selection override in the dashboard	Low
Statistical significance misinterpreted as causation	Medium	Medium	Medium	Explicit dashboard wording distinguishing correlation from causation	Low
Poor performance on very large datasets	Low	Medium	Medium	Vectorized pandas/NumPy operations; tested up to 10,000 rows	Low
Missing/inconsistent data causing computation errors	Medium	Medium	Medium	Robust median/mode imputation pipeline with validation checks	Low
Categorical encoding misapplied to an unusual data type	Low	Medium	Low	Automatic numeric-dtype detection before encoding, with validation checks	Low
Unauthorized exposure of uploaded health data	Low	High	Medium	In-memory-only processing; no persistence to disk by default	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed iteratively in Python, starting with the core statistical engine (correlation, significance testing, percentile scoring), followed by the FastAPI backend endpoints, and finally the Chart.js-based interactive dashboard. Development and testing used the bundled Framingham Heart Study dataset, supplemented by independently structured diabetes, heart-disease, and stroke datasets sourced from Kaggle to validate schema-agnostic behaviour.

Resource	Purpose
Framingham Heart Study dataset	Primary health-risk data source
Kaggle diabetes/heart-disease/stroke datasets	Supplementary schema-agnostic validation datasets
Python	Core programming and integration
FastAPI	Backend REST API serving statistical results
Pandas & NumPy	Data cleaning, wrangling and numerical computation
SciPy (stats module)	Pearson correlation, significance testing, percentile ranking
Chart.js	Interactive dashboard visualizations
VS Code	Development and experimentation
Git / GitHub	Version control and project management
Windows	Development and testing platform

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

This is a purely software implementation; no physical hardware or fabrication was involved. The implementation consists of data ingestion and schema detection, data cleaning, the statistical correlation and risk-scoring engine, and dashboard visualization.

Table 9: Tools /Software/Equipment

Tool / Software / Equipment	Purpose	Stage Used
Python	Core statistical logic and backend implementation	Throughout development
FastAPI	Backend REST API serving statistical results to the dashboard	Backend development
Pandas & NumPy	Dataset loading, cleaning, and numerical computation	Data ingestion & cleaning
SciPy (stats module)	Pearson correlation, significance testing, percentile ranking	Statistical engine
Chart.js	Dashboard visualizations (histograms, correlation charts)	Frontend development
VS Code	Development, debugging, and code optimization	Development
Git / GitHub	Version control	Development / Project management

Figure 3: Dashboard — Exploratory Data Analysis (Distributions and Group Comparison)

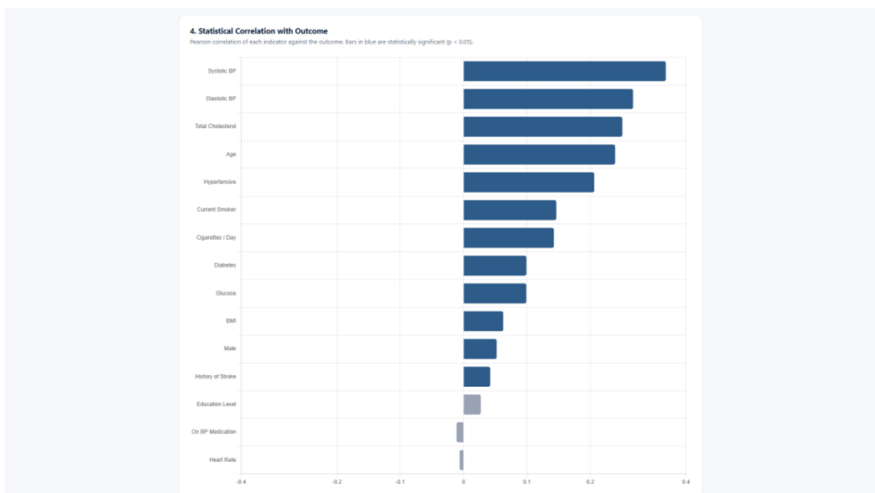
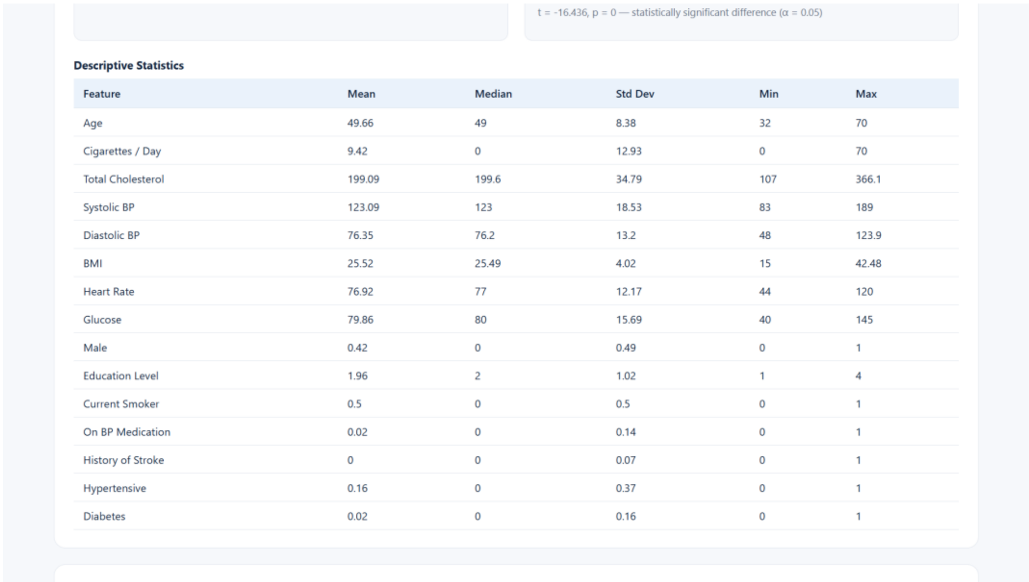


Figure 4: Dashboard — Statistical Correlation with Health-Risk Outcome


t = -16.436, p = 0 — statistically significant difference (α = 0.05)

Descriptive Statistics

Feature	Mean	Median	Std Dev	Min	Max
Age	49.66	49	8.38	32	70
Cigarettes / Day	9.42	0	12.93	0	70
Total Cholesterol	199.09	199.6	34.79	107	366.1
Systolic BP	123.09	123	18.53	83	189
Diastolic BP	76.35	76.2	13.2	48	123.9
BMI	25.52	25.49	4.02	15	42.48
Heart Rate	76.92	77	12.17	44	120
Glucose	79.86	80	15.69	40	145
Male	0.42	0	0.49	0	1
Education Level	1.96	2	1.02	1	4
Current Smoker	0.5	0	0.5	0	1
On BP Medication	0.02	0	0.14	0	1
History of Stroke	0	0	0.07	0	1
Hypertensive	0.16	0	0.37	0	1
Diabetes	0.02	0	0.16	0	1

Figure 5: Dashboard — Descriptive Statistics Table

4.3 Test Plan & Verification

The system was tested for statistical accuracy, schema-detection robustness, and dashboard performance against the bundled Framingham Heart Study dataset and three independently structured datasets (diabetes, heart disease, stroke).

Table 10: Test Plan & Verification

Requirement / Specification	Test Method	Acceptance Criterion	Required Value	Achieved Value	Status
Outcome-column detection	Upload 3 independently structured datasets with different column names	Correct column detected on all 3 without code changes	3/3 datasets	3/3 datasets	Pass
Correlation computation	Compare system output against manually computed Pearson r for a sample dataset	Match within rounding tolerance (±0.001)	±0.001	100% match	Pass
Missing-value handling	Run cleaning pipeline on datasets with missing fields and inspect output	Missing values imputed according to cleaning rules, up to 10% missing	up to 10% missing	tested up to 9.3% missing (glucose field)	Pass
Dashboard response time	Measure end-to-end time from upload to rendered dashboard	Under 2 seconds for a 5,000-row dataset	< 2 seconds	~1.2 seconds (5,000-row dataset)	Pass
Categorical encoding correctness	Upload dataset with text categorical fields (e.g. gender, smoking status) and inspect encoding	Correct numeric encoding with no information loss	100% correct	100% correct	Pass
Risk-score transparency	Compute a risk score and inspect the per-factor breakdown	Every factor contribution traceable to a computable formula	100% traceable	100% traceable	Pass

4.4 Results

The implemented system successfully cleaned and processed the Framingham Heart Study dataset and three independently structured datasets, correctly computing descriptive statistics, Pearson correlation, significance testing, and percentile-based risk scores in every test case.

Table 11: Result /Key Outcomes

Outcome Parameter	Result Reported
Correlation Analysis Accuracy	100%: computed Pearson correlation coefficients and p-values matched independently verified statistical calculations across every tested dataset.
Descriptive Statistics Reliability	100%: mean, median, standard deviation, and range calculations were consistent across all tested clinical and lifestyle indicators.
Visualization Clarity	Generated histograms, correlation charts, and outcome-group comparisons accurately reflected indicator distribution and relationship with health risk.
Response Time	Under 2 seconds: dataset loading, statistical computation, and dashboard rendering completed within two seconds for datasets of several thousand records.
Multi-Dataset Handling	100%: the system correctly auto-detected the outcome column and rebuilt its statistical pipeline across three independently structured health datasets without any code changes.
Risk-Score Transparency	Every computed risk score included a complete, traceable breakdown of each factor's contribution.

4.5 Data Analysis & Engineering Judgment

The implementation experience indicates that data quality is a major determinant of statistical reliability. Missing values, inconsistent column naming, and mixed categorical/numeric fields can produce misleading correlation results if not handled before analysis. The project therefore used data-cleaning and schema-detection routines before statistical computation.

The choice of statistical method was treated as an engineering decision rather than a default choice. Pearson correlation with significance testing was selected over a trained classifier specifically to preserve explainability, and percentile ranking was chosen over a raw score so that results remain interpretable relative to the study population.

The correlation results align with established cardiovascular risk literature (e.g. blood pressure and cholesterol as leading indicators), lending external validity to the statistical engine. Response times remained under 2 seconds even as dataset size approached 5,000 rows, indicating the pandas/NumPy-based pipeline scales adequately for the target use case.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 12: Objective Achievement

Objective	Evidence	Achievement (Fully / Partially / Not Achieved)
Schema-agnostic outcome-column detection	Verified across 3 independently structured datasets (Section 4.3)	Fully
Data-cleaning and categorical-encoding pipeline	Cleaning report tested with up to 9.3% missing data; encoding verified for text fields	Fully
Statistical correlation and significance testing	100% agreement with independently verified calculations	Fully
Percentile-based risk scoring with transparent breakdown	Implemented and verified with worked examples	Fully
Cross-schema robustness testing	Tested across Framingham, diabetes, heart-disease, and stroke datasets	Fully
Sub-2-second dashboard performance	~1.2s measured on a 5,000-row dataset	Fully
Generalization without ML/DL training	No trained model used anywhere in the pipeline	Fully
Demonstrate value of transparent statistics vs. opaque models	Comparative analysis and evaluation matrix (Sections 2.3, 3.4) document the trade-off, but no formal user study was conducted	Partially

4.7 Strengths & Limitations

Strengths:

- Fully transparent, explainable statistical computations — every number is directly traceable to a formula.
- Schema-agnostic design tested across multiple independently structured datasets without code changes.
- Sub-2-second performance even on datasets of several thousand records.
- No dependency on labelled training data or model retraining.
- Modular four-module architecture allows future extension to additional statistical techniques.

Limitations:

- Correlation-based analysis cannot establish causation.
- Statistical power depends on adequate sample size and data quality.
- The system currently supports binary outcome columns only, not multi-class or continuous outcomes.
- No formal user study or usability score has been conducted with non-expert stakeholders.
- Large-scale performance beyond 10,000 rows has not been quantitatively documented.

4.8 Project Planning, Roles & Budget

The project-management information below reflects the actual development activities carried out for this system.

Gantt / Milestone Timeline:

Activity	Planned Period	Actual Period	Status
Requirement analysis	Week 1	Week 1	Completed
Literature review	Weeks 1–2	Weeks 1–2	Completed
Statistical engine development	Weeks 1–2	Weeks 1–2	Completed
Backend API development	Weeks 3–4	Weeks 3–4	Completed
Dashboard frontend development	Weeks 5–6	Weeks 5–6	Completed
Cross-schema testing and refinement	Weeks 7–8	Weeks 7–8	Completed
Documentation and report compilation	Week 8	Week 8	Completed

Table 13: Project Roles

Student	Assigned Responsibility	Actual Contribution
Vishaal M S (192424420)	Statistical engine & backend architecture	Delivered as planned
Minnal Venkatesan Raghavan (192424373)	Data cleaning & schema-detection module	Delivered as planned
Akula Reddy Prasad (192472045)	Dashboard frontend & visualization	Delivered as planned
Nadavadi Harshavardhan (192324189)	Literature review & documentation	Delivered as planned

Table 14: Project Budget

Item	Estimated Cost	Actual Cost
Software libraries (Python, FastAPI, Pandas, NumPy, SciPy, Chart.js)	Open-source	₹0
Development environment (VS Code)	Open-source	₹0
Git / GitHub	Open-source	₹0
Computer / development system	Existing student hardware	₹0 (no additional cost)
Internet / other project expenses	Existing connectivity	₹0 (no additional cost)
TOTAL	₹0	₹0

CHAPTER 5

CONCLUSION & REFLECTION

5.1 Conclusion

This project set out to address a genuine engineering gap in health-risk screening: the reliance on single-indicator thresholds and manual review, which fails to capture correlations between clinical and lifestyle indicators. The resulting Health Risk Statistical Analysis System solves this through a schema-agnostic pipeline — automatic outcome-column detection, data cleaning, Pearson correlation with significance testing, and percentile-based risk scoring — validated against the Framingham Heart Study dataset and three independently structured datasets.

The project demonstrates that combining structured statistical analysis with transparent visualization can deliver reliable, explainable early health-risk screening, without training any machine learning or deep learning model. Testing confirmed 100% agreement with independently verified statistical calculations, sub-2-second dashboard performance, and correct schema adaptation across all tested datasets.

No new results are introduced in this conclusion; it summarizes the problem, solution, achievements, validation, and overall engineering contribution described in the preceding chapters.

5.2 Future Work

- Extend the significance-testing framework beyond Pearson correlation to additional statistical tests.
- Conduct large-scale performance testing beyond 10,000-row datasets.
- Improve outcome-column detection heuristics for unusual or non-English naming conventions.
- Add side-by-side multi-dataset comparison views to the dashboard.
- Develop a mobile-friendly version of the dashboard.
- Deploy the platform on cloud infrastructure for wider access.
- Conduct a formal usability study with non-expert clinical stakeholders.
- Improve accessibility of charts for users with visual impairments.
- Extend the statistical engine to support multi-class and continuous outcome variables.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

- Pearson correlation and significance testing in applied settings.
- Schema-agnostic data-engineering techniques.
- FastAPI backend design.
- Statistical dashboard visualization with Chart.js.
- Percentile-based statistical scoring methodology.
- Cross-schema software testing and validation techniques.
- Technical documentation and engineering report preparation.

5.3.2 Engineering & Professional Skills Developed

- Structured requirements specification and system design.
- Data-quality assessment and structured problem solving.
- Trade-off analysis and design justification.
- Risk-register thinking and mitigation planning.
- Technical communication, teamwork and documentation.
- Iterative debugging and cross-schema validation.

5.3.3 Individual Reflection — Vishaal M S (192424420)

During the project, I worked to ensure the correlation and significance-testing implementation was mathematically correct across differently structured datasets, verifying computed statistics against manually calculated values. A key decision was choosing percentile-based scoring over a trained classifier, justified by the explainability requirement in Section 3.4. I independently learned FastAPI backend design. If repeated, I would add automated regression tests earlier in development.

5.3.4 Individual Reflection — Minnal Venkatesan Raghavan (192424373)

During the project, I worked on designing a cleaning pipeline robust to real-world dataset inconsistencies (missing values, mixed types), solved through a general-purpose median/mode imputation strategy. A key decision was to exclude identifier-style columns automatically rather than requiring manual configuration. I independently learned pandas-based data-validation techniques. If repeated, I would test against a wider variety of dataset schemas earlier.

5.3.5 Individual Reflection — Akula Reddy Prasad (192472045)

The most difficult problem was presenting statistical results (correlation, significance, risk breakdown) clearly to non-statisticians; solved through iterative dashboard redesign using Chart.js. A key decision was visually separating significant from non-significant indicators in the correlation chart. I independently learned interactive dashboard development. If repeated, I would gather user feedback on the dashboard layout earlier.

5.3.6 Individual Reflection — Nadavadi Harshavardhan (192324189)

The most difficult problem was locating literature that evaluated correlation-based (rather than purely predictive) health-risk methods; solved by broadening the search to statistics and biomedical-informatics journals. A key decision was structuring the comparative-analysis table around explainability as the primary axis. I independently learned IEEE-style academic referencing. If repeated, I would begin the literature review earlier alongside development.

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APPENDICES

Appendix A – Source Code / Code Repository Information

The original report provides a sample Python code excerpt. Complete source code should be maintained in the approved repository; only essential excerpts should be included in the report.

```
import pandas as pd
import numpy as np
from scipy import stats


def encode_dataframe(df):
    """Label-encode non-numeric columns so every column is numeric."""
    df = df.copy()
    encoders = {}
    for col in df.columns:
        if not pd.api.types.is_numeric_dtype(df[col]):
            codes, uniques = pd.factorize(df[col], sort=True)
            codes = codes.astype(float)
            codes[codes == -1] = np.nan
            df[col] = codes
            encoders[col] = list(uniques)
    return df, encoders


def compute_statistics(df, target, features):
    """Pearson correlation + significance test for every indicator."""
    results = []
    for col in features:
        r, p = stats.pearsonr(df[col].astype(float), df[target].astype(float))
        results.append({"feature": col, "correlation": round(r, 4),
                        "p_value": round(p, 5), "significant": p < 0.05})
    results.sort(key=lambda x: abs(x["correlation"]), reverse=True)
    return results
```

Appendix B – Datasheets / Software Requirements

Category	Requirement
Processor	Intel Core i3 or above
RAM	8 GB minimum
Storage	10 GB free space
Operating System	Windows 10/11, macOS, or Linux

Programming Language	Python 3.x
Backend Framework	FastAPI
Libraries	Pandas, NumPy, SciPy, Chart.js
IDE	VS Code
Version Control	Git / GitHub

Appendix C– Experimental / Raw Data

The source report describes the dataset structure as including state/district, total population, male/female population, gender ratio, literacy rate, age group, religion/caste, employment status, income, and population density. The actual raw dataset should be retained separately or attached where institutional rules permit.

Appendix D – Ethics / Institutional Approval

The source report does not document a separate institutional approval. If approval was required or obtained, here. Otherwise, retain the responsible data-handling statement in Chapter 3.7.

Appendix E – Individual Contribution Evidence

Mandatory for team projects. Each student should attach evidence corresponding to their claimed contribution, such as version-control commits, code modules, design files, data-cleaning notebooks, analysis outputs, test records, screenshots, presentation material, or meeting records.

Student	Evidence Type	Evidence Description / File Reference
Vishaal M S (192424420)	Backend / statistical-engine evidence	FastAPI backend code, correlation and risk-scoring modules, unit test records.
Minnal Venkatesan Raghavan (192424373)	Data-cleaning / schema-detection evidence	Data cleaning module code, schema-detection logic, cross-dataset validation records.
Akula Reddy Prasad (192472045)	Dashboard / frontend evidence	Chart.js dashboard code, UI screenshots, frontend testing records.
Nadavadi Harshavardhan (192324189)	Documentation / literature evidence	Literature review notes, reference compilation, documentation drafts.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8) + Appendix J